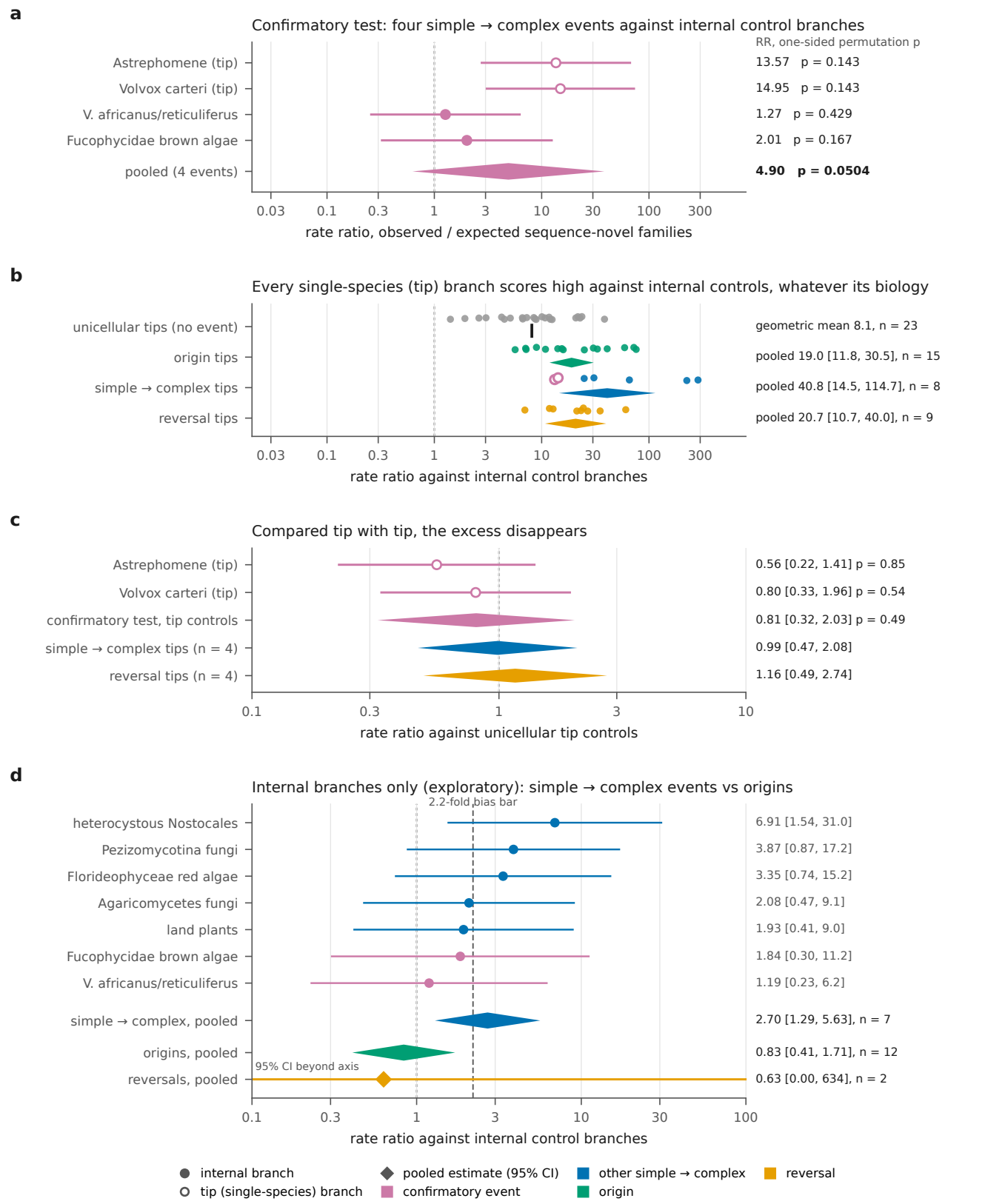
a
Confirmatory test: four simple → complex events against internal control branches
RR, one-sided permutation p
Astrephomene (tip)
Volvox carteri (tip)
V. africanus/reticuliferus
Fucophycidae brown algae
pooled (4 events)
13.57 p = 0.143
14.95 p = 0.143
1.27 p = 0.429
2.01 p = 0.167
4.90 p = 0.0504
0.03 0.1 0.3 1 3 10 30 100 300
rate ratio, observed / expected sequence-novel families
b
Every single-species (tip) branch scores high against internal controls, whatever its biology
unicellular tips (no event)
origin tips
simple → complex tips
reversal tips
geometric mean 8.1, n = 23
pooled 19.0 [11.8, 30.5], n = 15
pooled 40.8 [14.5, 114.7], n = 8
pooled 20.7 [10.7, 40.0], n = 9
0.03 0.1 0.3 1 3 10 30 100 300
rate ratio against internal control branches
c
Compared tip with tip, the excess disappears
Astrephomene (tip)
Volvox carteri (tip)
confirmatory test, tip controls
simple → complex tips (n = 4)
reversal tips (n = 4)
0.56 [0.22, 1.41] p = 0.85
0.80 [0.33, 1.96] p = 0.54
0.81 [0.32, 2.03] p = 0.49
0.99 [0.47, 2.08]
1.16 [0.49, 2.74]
0.1 0.3 1 3 10
rate ratio against unicellular tip controls
d
Internal branches only (exploratory): simple → complex events vs origins
2.2-fold bias bar
heterocystous Nostocales
Pezizomycotina fungi
Florideophyceae red algae
Agaricomycetes fungi
land plants
Fucophycidae brown algae
V. africanus/reticuliferus
simple → complex, pooled
origins, pooled
95% CI beyond axis
reversals, pooled
6.91 [1.54, 31.0]
3.87 [0.87, 17.2]
3.35 [0.74, 15.2]
2.08 [0.47, 9.1]
1.93 [0.41, 9.0]
1.84 [0.30, 11.2]
1.19 [0.23, 6.2]
2.70 [1.29, 5.63], n = 7
0.83 [0.41, 1.71], n = 12
0.63 [0.00, 634], n = 2
0.1 0.3 1 3 10 30 100
rate ratio against internal control branches
internal branch
tip (single-species) branch
pooled estimate (95% CI)
confirmatory event
other simple → complex
origin
reversal